

✈️ AVIATION DATA ANALYST

Portfolio Project Worksheet

3 Projects | Easy → Medium → Hard | Real Aviation Data

Instructions

- Complete all 3 projects in order — each one builds on the previous.
- Use real, free data sources linked in each project.
- Upload your code and output files to a GitHub repository.
- Answer all reflection questions in the spaces provided.
- All three projects together form your data analyst portfolio.

PROJECT 1 OF 3

● Airline On-Time Performance Dashboard

DIFFICULTY
EASY

ESTIMATED TIME
1 – 2 Weeks

TOOLS
Excel / Python + Tableau or Power BI

Overview

You will download real US airline delay data, clean it, and build a dashboard that answers key operational questions. This mirrors exactly what airline performance analysts do every day.

Data Source

BTS On-Time Performance Data → transtats.bts.gov/OT_Delay/OT_DelayCause1.asp

- Download at least 3 months of data as a CSV file.
- It is completely free — no account needed.

Tasks

Step 1 — Load & Inspect the Data

1. Open the CSV in Excel or load it into a Python notebook using pandas.
2. How many rows and columns does the dataset have? Write below.

Step 2 — Clean the Data

3. Remove rows where the carrier name or airport code is missing.
4. Check for duplicate rows and remove them.
5. Rename at least 3 columns to make them more readable.
6. List the 3 columns you renamed and why:

Step 3 — Analyze the Data

Answer each question using pivot tables (Excel) or groupby (Python):

7. Which airline has the worst on-time performance? (% of delayed flights)
8. What is the most common delay cause? (weather / carrier / NAS / security / late aircraft)
9. Which 5 routes have the most delays?
10. Is there a day of week or time of year with more delays?

Step 4 — Build the Dashboard

11. Create a dashboard in Tableau Public or Power BI with at least 3 charts.
12. Publish it online (Tableau Public is free) and paste the link below.

Dashboard Link:

Deliverables Checklist

- Cleaned dataset (CSV or Excel file) uploaded to GitHub
- Python notebook or Excel file showing your analysis
- Published dashboard (Tableau Public or Power BI link)
- All reflection questions answered below

Reflection Questions

Q1. What surprised you most about the delay data?

Q2. If you were an airline operations manager, what one change would you make based on your findings?

Q3. What was the hardest part of cleaning this data?

PROJECT 2 OF 3

● Flight Route & Traffic Analysis (Live ADS-B Data)

DIFFICULTY
MEDIUM

ESTIMATED TIME
2 – 3 Weeks

TOOLS
Python (pandas, Folium, requests)

Overview

You will connect to the OpenSky Network API to pull real live flight tracking data, analyze traffic patterns, and visualize routes on an interactive map. This is the type of work done in Air Traffic Management and airline network planning.

Data Source

OpenSky Network → opensky-network.org (free account required)

- Create a free account at opensky-network.org.
- You will access data using Python and the `opensky-api` library.
- Install with: `pip install opensky-api` or `pip install requests`

Setup Task

Before you begin, run the following Python code to confirm your API connection works:

```
import requests
r = requests.get('https://opensky-network.org/api/states/all')
print(r.status_code) # should print 200
```

Tasks

Step 1 — Pull & Understand the Data

13. Call the OpenSky API and save the response to a pandas DataFrame.
14. List the column names you receive. What does each one mean?

Step 2 — Clean & Filter

15. Remove rows with missing latitude, longitude, or callsign.
16. Filter to flights only over a region of your choice (e.g. Europe, USA, Gulf).
17. How many flights remain after filtering? Write below.

Step 3 — Analyze Traffic Patterns

18. What are the 10 busiest departure airports in your filtered dataset?
19. What is the average altitude of flights in your region?
20. Are there more flights in certain hours of the day? (Use the 'time_position' column)
21. Which country has the most registered aircraft in your dataset?

Step 4 — Map the Routes

22. Install Folium: `pip install folium`
23. Plot each flight as a marker on an interactive HTML map.
24. Color the markers by altitude (low = red, medium = orange, high = green).
25. Save the map as an HTML file and upload it to your GitHub repo.

GitHub link to your HTML map file:

Deliverables Checklist

- Python notebook (.ipynb) with all analysis uploaded to GitHub
- Cleaned CSV or parquet file of your filtered flight data
- Interactive HTML route map published on GitHub
- All reflection questions answered

Reflection Questions

Q1. How does real ADS-B data differ from the structured BTS data in Project 1? What challenges did it bring?

Q2. Based on traffic density, which airport or corridor surprised you most and why?

Q3. How could an airline or ANSP use this type of analysis in real operations?

PROJECT 3 OF 3

Flight Delay Prediction Model (Machine Learning)

DIFFICULTY
HARD

ESTIMATED TIME
3 – 4 Weeks

TOOLS
Python (pandas,
scikit-learn, matplotlib)

Overview

You will build a machine learning model that predicts whether a flight will be delayed based on route, airline, time of day, and weather. This is the most advanced project and ties everything together. Airlines and airports use similar models for resource planning and passenger communication.

Data Sources

1. **BTS On-Time Data** → transtats.bts.gov (same as Project 1)

2. **NOAA Weather Data** → ncdc.noaa.gov/cdo-web (free account)

- From NOAA, download airport weather records (wind speed, visibility, precipitation) for the same airports and time period as your BTS data.
- You will need to merge the two datasets by airport code and date.

Tasks

Step 1 — Prepare & Merge the Data

26. Load both datasets into pandas DataFrames.

27. Create a shared key to merge them. What columns did you use to join? Write below.

28. After merging, how many rows does the final dataset have?

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Step 2 — Feature Engineering

Feature engineering = creating new columns from existing data that help the model learn.

29. Create a column: 'is_delayed' (1 = delayed more than 15 minutes, 0 = not delayed).
30. Extract from the departure time: hour of day, day of week, month, season.
31. Create a 'route' column by combining origin + destination airport codes.

32. Which weather columns do you think will be most useful for predicting delays?
Why?

Step 3 — Build the Model

- 33. Split your data: 80% training, 20% testing.
- 34. Train a Random Forest classifier using scikit-learn on the training set.
- 35. Predict delay outcomes on the test set.
- 36. Paste your model accuracy score below:

Step 4 — Evaluate & Interpret

- 37. Generate a confusion matrix. What do the 4 cells mean in plain English?

- 38. Which feature (column) had the highest importance score in your Random Forest?

- 39. Would you trust this model to make real decisions? What are its limitations?

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Step 5 — Simple Predictor

40. Write a Python function that takes: airline, origin airport, destination airport, departure hour, and a weather condition — and returns a predicted delay risk (High / Medium / Low).
41. Test it on 3 different example flights and record the results below.

Deliverables Checklist

- Full Python notebook with data prep, model training, and evaluation
- Confusion matrix chart saved as an image
- Feature importance chart saved as an image
- One-page PDF summary of your findings (what the model does, accuracy, limitations)
- Everything uploaded to a public GitHub repository

Reflection Questions

Q1. What was the single most important predictor of flight delays in your model?

Q2. If you were presenting this model to an airline operations team, what 3 things would you tell them about its limitations?

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Q3. How would you improve this model if you had 6 more months and access to more data?



Portfolio Complete!

Once all 3 projects are on GitHub, you have a job-ready data analyst portfolio with real aviation domain knowledge that most candidates don't have.

Target Roles: Aviation Data Analyst | Airline Operations Analyst | Flight Data Analyst